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Task 1

```
C# Program.cs
1  using System;
2
3  class Program
4  {
5      public static void Main()
6      {
7
8          string userName = "Ram";
9          int luckyNumber = 7;
10         Console.WriteLine($"The name is {userName}");
11     }
12 }
13
```

PROBLEMS OUTPUT TERMINAL ...  zsh     ...  

```
● rakeshgurung@Rakeshs-MacBook-Air Practice % dotnet run
The name is Ram! Your lucky number is 7
○ rakeshgurung@Rakeshs-MacBook-Air Practice %
```

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Task 2

```
C# circle.cs
1  class Circle
2  {
3      // Constant value of PI – cannot be changed
4      public const double PI = 3.14;
5
6      public double Radius { get; set; }
7
8      // Constructor
9      public Circle(double radius)
10     {
11         Radius = radius;
12     }
13
14     public double CalculateArea()
15     {
16         return PI * Radius * Radius;
17     }
18
19     // Method to calculate perimeter (circumference)
20     public double CalculatePerimeter()
21     {
22         return 2 * PI * Radius;
23     }
24 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh + - [] [] ... ^ x

○ rakeshgurung@Rakeshs-MacBook-Air Practice % []

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C# Program.cs

```
1  using System;
2
3  class Program
4  {
5      public static void Main()
6      {
7
8          Circle c1 = new Circle(5);
9
10         Console.WriteLine($"Radius = {c1.Radius}");
11         Console.WriteLine($"Area = {c1.CalculateArea()}");
12         Console.WriteLine($"Perimeter = {c1.CalculatePerimeter()}");
13     }
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

zsh + ▢

```
● rakeshgurung@Rakeshs-MacBook-Air Practice % dotnet run
Radius = 5
Area = 78.5
Perimeter = 31.400000000000002
○ rakeshgurung@Rakeshs-MacBook-Air Practice %
```

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Task 3

```
C# Program.cs
5 public static void Main()
6
7     // Declare and initialize variables
8     byte byteVar = 255;
9     short shortVar = -32768;
10    int intVar = 1000;
11    long longVar = 100000L;
12    float floatVar = 3.14f;
13    double doubleVar = 3.14159;
14    decimal decimalVar = 123.456m;
15    char charVar = 'A';
16    bool boolVar = true;
17
18    // Convert integer value 42 to a string
19    int number = 42;
20    string stringNumber = number.ToString();
21
22    // Convert string "3.14" to a double
23    string strValue = "3.14";
24    double parsedDouble = Convert.ToDouble(strValue);
25
26    Console.WriteLine($"byteVar (byte): {byteVar}");
27    Console.WriteLine($"shortVar (short): {shortVar}");
28    Console.WriteLine($"intVar (int): {intVar}");
29    Console.WriteLine($"longVar (long): {longVar}");
30    Console.WriteLine($"floatVar (float): {floatVar}");
31    Console.WriteLine($"doubleVar (double): {doubleVar}");
32    Console.WriteLine($"decimalVar (decimal): {decimalVar}");
33    Console.WriteLine($"charVar (char): {charVar}");
34    Console.WriteLine($"boolVar (bool): {boolVar}");
35    Console.WriteLine($"stringNumber (converted from int 42): {stringNumber}");
36    Console.WriteLine($"parsedDouble (converted from \"3.14\"): {parsedDouble}");
37
```

● rakeshgurung@Rakeshs-MacBook-Air Practice % dotnet run

```
byteVar (byte): 255
shortVar (short): -32768
intVar (int): 1000
longVar (long): 100000
floatVar (float): 3.14
doubleVar (double): 3.14159
decimalVar (decimal): 123.456
charVar (char): A
boolVar (bool): True
stringNumber (converted from int 42): 42
parsedDouble (converted from "3.14"): 3.14
```

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Task 4

```
C# Program.cs
7 // 1. Creating array
8 int[] favoriteNumbers = { 7, 3, 42, 15, 9 };
9
10 Console.WriteLine("Original array:");
11 for (int i = 0; i < favoriteNumbers.Length; i++)
12 {
13     Console.Write(favoriteNumbers[i] + " ");
14 }
15 Console.WriteLine();
16
17 // 2. Sorting array in ascending order
18 Array.Sort(favoriteNumbers);
19 Console.WriteLine("\nArray after sorting (ascending):");
20 for (int i = 0; i < favoriteNumbers.Length; i++)
21 {
22     Console.Write(favoriteNumbers[i] + " ");
23 }
24 Console.WriteLine();
25
26 // 3. Reverse the sorted array
27 Array.Reverse(favoriteNumbers);
28 Console.WriteLine("\nArray after reversing (descending):");
29 for (int i = 0; i < favoriteNumbers.Length; i++)
30 {
31     Console.Write(favoriteNumbers[i] + " ");
32 }
33 Console.WriteLine();
34
35 // 4. Finding the position
36 int numberToFind = 15;
37 int index = Array.IndexOf(favoriteNumbers, numberToFind);
38
39 if (index != -1)
40 {
41     Console.WriteLine($"The number {numberToFind} is at index {index} in the reverse
42 }
43 else
```

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```
43         else
44         {
45             Console.WriteLine($"\\nThe number {numberToFind} is not in the array.");
46         }
47     }
48 }
```

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```
• rakeshgurung@Rakeshs-MacBook-Air Practice % dotnet run
Original array:
7 3 42 15 9

Array after sorting (ascending):
3 7 9 15 42

Array after reversing (descending):
42 15 9 7 3

The number 15 is at index 1 in the reversed array.
○ rakeshgurung@Rakeshs-MacBook-Air Practice %
```

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Task 5

```
Program.cs
1  using System;
2
3  class Program
4  {
5      public static void Main(){
6          // Creating DateTime variable
7          DateTime birthDate = new DateTime(1986, 5, 15);
8
9          // Creating DateTime variable for the current date and time
10         DateTime currentDate = DateTime.Now;
11
12         // Calculating the age using TimeSpan
13         TimeSpan ageSpan = currentDate - birthDate;
14         int ageInYears = (int)(ageSpan.Days / 365.25); // Approximate age in years
15
16         Console.WriteLine($"Birthdate: {birthDate.ToShortDateString()}");
17         Console.WriteLine($"Current Date: {currentDate.ToShortDateString()}");
18         Console.WriteLine($"Age: {ageInYears} years");
19
20         // Add 10 days to birthdate
21         DateTime newDate = birthDate.AddDays(10);
22         Console.WriteLine($"Birthdate + 10 days: {newDate.ToShortDateString()}");
23     }
24 }
25
```

Ports

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- rakeshgurung@Rakeshs-MacBook-Air Practice % dotnet run
Birthdate: 15/05/1986
Current Date: 04/11/2025
Age: 39 years
Birthdate + 10 days: 25/05/1986
- rakeshgurung@Rakeshs-MacBook-Air Practice % █

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Task 6

```
Program.cs
3  class Program
4  {
5      public static void Main(){
6
7          List<string> fruits = new List<string> { "Apple", "Banana", "Mango" };
8
9          // Adding a new fruit in array
10         fruits.Add("Orange");
11
12         // Removing one fruit
13         fruits.Remove("Banana");
14
15         // Printing all fruits using a foreach loop
16         Console.WriteLine("Fruits in the list:");
17         foreach (string fruit in fruits)
18         {
19             Console.WriteLine(fruit);
20         }
21
22         // 5. Creating a Dictionary with fruit IDs and names
23         Dictionary<int, string> fruitDictionary = new Dictionary<int, string>
24         {
25             { 1, "Apple" },
26             { 2, "Banana" },
27             { 3, "Mango" }
28         };
29     }
```


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```
29
30      // 6. Adding a new entry to the dictionary
31      fruitDictionary.Add(4, "Orange");
32
33      // 7. Printing all key-value pairs in the dictionary
34      Console.WriteLine("\nFruit Dictionary:");
35      foreach (KeyValuePair<int, string> kvp in fruitDictionary)
36      {
37          Console.WriteLine($"ID: {kvp.Key}, Name: {kvp.Value}");
38      }
39  }
40  }
41
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Fruits in the list:
Apple
Mango
Orange

Fruit Dictionary:
ID: 1, Name: Apple
ID: 2, Name: Banana
ID: 3, Name: Mango
ID: 4, Name: Orange
rakeshgurung@Rakeshs-MacBook-Air Practice %